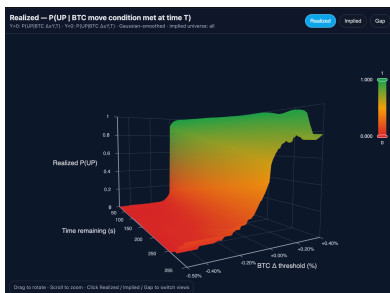


Do Prediction Markets Price Reality?

Exploring calibration in Polymarket's Bitcoin 5-minute markets

COM-480 Data Visualization Process Book

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Final visualization: an interactive 3D calibration surface comparing market-implied probabilities with realized Bitcoin outcomes.

Introduction

This project explores whether short-term prediction markets behave like reliable forecasting systems. We focus on Polymarket's Bitcoin 5-minute markets, where traders buy and sell tokens depending on whether Bitcoin will be higher or lower at the end of a five-minute window.

To study this, we built a custom dataset combining Polymarket market data with Binance Bitcoin spot prices. Our final website turns this data into an interactive visual story: users first understand the market mechanics and historical patterns, then explore calibration through a 3D visualization, and finally test their intuition in a simplified trading sandbox based on past market data.

Motivation and Data Story

Prediction markets such as Polymarket have gained increasing attention because they transform uncertain future events into prices. In theory, these prices can be interpreted as probabilities: if a token trades at 70 cents, the market is effectively saying that the event has around a 70% chance of happening.

This idea is closely related to the classic “wisdom of the crowd” phenomenon. A common example is the village fair experiment, where many people were asked to guess the weight of an ox. Although individual guesses varied widely, the average guess was surprisingly close to the true weight. Prediction markets follow a similar intuition: by aggregating the beliefs of many participants, especially when real money is involved, the market price may become a useful estimate of reality.

However, our project asks whether this intuition still holds in a much faster and noisier setting. Instead of long-term political or sports events, we focus on Polymarket's Bitcoin 5-minute markets. In each market, traders bet on whether Bitcoin will be higher or lower at the end of a five-minute window. These markets are repeated throughout the day, creating hundreds of comparable events and a large high-frequency dataset of prices, trades, Bitcoin movements, and final outcomes.

Our main question is: **do market-implied probabilities actually match historical outcomes?** For example, when the UP token is priced at 70 cents, does Bitcoin really finish up around 70% of the time? More importantly, are there situations where the crowd systematically overestimates or underestimates the probability of an outcome?

The data story we tell is therefore about calibration. We do not simply ask whether the market is “right” or “wrong” in individual cases. Instead, we examine when the market behaves like a well-calibrated forecasting system, and when it appears biased or uncertain. We explore how this changes depending on time remaining, Bitcoin price movement, and the implied probability shown by the market.

Our final website first guides users through the intuition behind prediction markets and presents historical insights from the dataset. It then leads to an interactive 3D visualization where users can explore the calibration surface themselves. Finally, we include a simplified trading sandbox based on past market data, allowing users to experience the difficulty of making short-term predictions in a simulated environment. The goal is to make a complex financial question understandable, visual, and engaging, while making clear that the project is educational and does not encourage gambling or provide trading advice.

Dataset

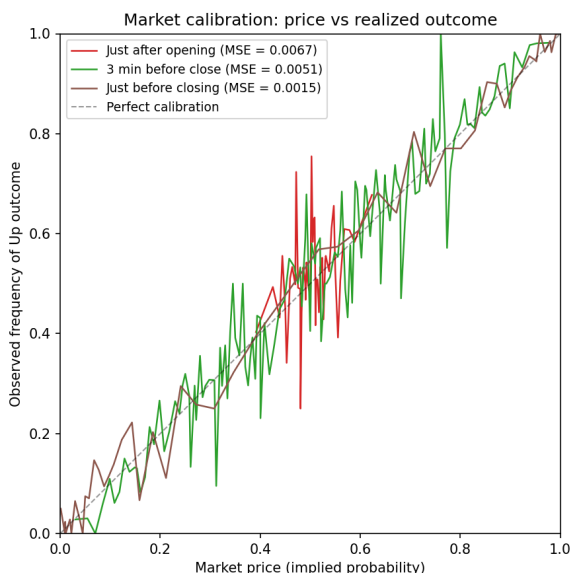
Our dataset was built specifically for this project, since Polymarket’s Bitcoin 5-minute markets do not exist as a ready-to-use research dataset. We queried Polymarket’s API to collect market metadata, outcomes, and trade sequences, then matched each round with Binance BTC spot prices. Each five-minute market has two tokens, **UP** and **DOWN**, whose prices can be interpreted as the market-implied probability of each outcome.

The dataset spans **February 12 to March 15, 2026** and contains **9,181 resolved markets**, representing around **16.8 million trades** and **\$686 million** in total volume. The overall UP rate is close to balanced, with **51.4%** of markets resolving UP and **48.6%** resolving DOWN. This scale makes the dataset well suited for studying calibration across many repeated and comparable events.

Our processing pipeline produced two main files: a market-level summary table with one row per market, and a per-second time series file with 300 observations per market. Before visualization, we aligned Polymarket and Binance timestamps, removed anomalous rounds, filtered negligible trades below 10 tokens, and computed derived variables such as time remaining and Bitcoin price change since market open.

Early Analysis and Milestone 1

Our first milestone was mainly exploratory. At that stage, our goal was not yet to build a polished story for a broad audience, but to understand whether the dataset contained meaningful patterns at all. We were still in a technical mindset: checking data quality, testing whether market prices could be compared to realized outcomes, and looking for signals that would justify a more ambitious visualization.



Last traded UP token price before resolution, separated by actual outcome.

Early calibration plot comparing implied probability with realized UP frequency.

The calibration plot already captured the central question of the project: if the market price is close to 0.7, does the event happen about 70% of the time? It suggested that the market becomes more calibrated as the round approaches its end. The last-trade distribution was useful as a sanity check: most markets ended with prices very close to 0 or 1, as expected when uncertainty almost disappears near the final seconds.

However, these early visualizations were designed more for analysis than communication. They were dense, static, and required prior knowledge of prediction markets, calibration, and statistical evaluation. Terms such as MSE were useful during analysis, but added clutter for a viewer who simply wanted to understand the story.

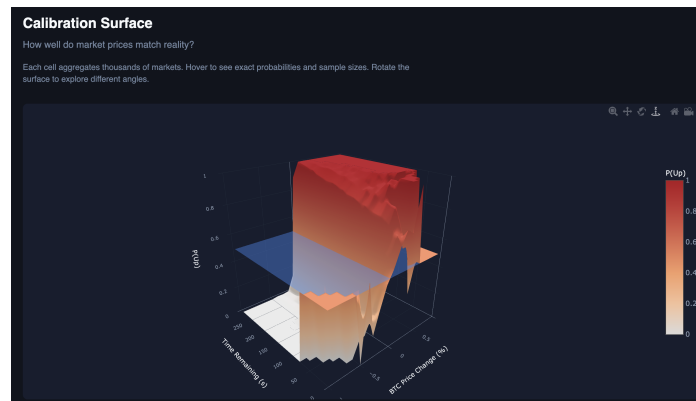
This revealed a major design challenge for the rest of the project. We did not only need to show that the market was sometimes calibrated or biased; we needed to make that idea understandable to someone seeing the topic for the first time. From this point onward, the project shifted from “showing results” to building a guided data story.

Milestone 2 Prototype

After Milestone 1, we moved from an investigation-oriented approach toward a more designed and understandable prototype. The goal was no longer only to prove that the data contained interesting patterns, but to shape those patterns into a visual experience that could work for different audiences: technical users looking for calibration gaps, and non-experts who first needed to understand what the market was doing.

The main design change was the introduction of a single-scroll narrative website. Instead of showing isolated analysis plots, the prototype organized the project into connected blocks: an introductory hero section, a

central 3D calibration surface, supporting distribution and heatmap views, and a Markov simulation for outcome sequences.



Milestone 2 prototype: an early 3D calibration surface showing how implied probability changes with time remaining and Bitcoin price movement.

The 3D surface became the visual centerpiece of the prototype. It translated the central question into one interactive object: how does the probability of an UP outcome vary with time remaining and BTC price change? Users could rotate and zoom the surface, which made the visualization feel closer to an exploratory financial tool than to a static academic plot.

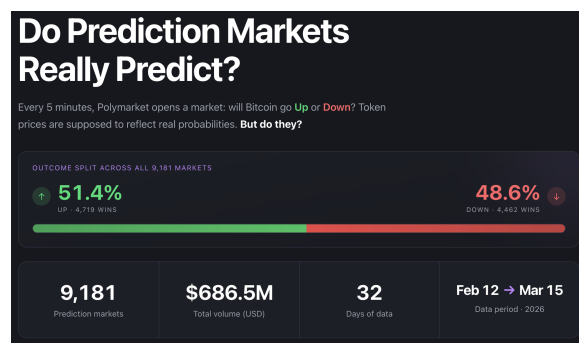
We also started using color more intentionally. In the early analysis, color mostly separated outcomes or curves. In the prototype, color became a storytelling tool: green and red represented UP and DOWN, sequential colors highlighted volume intensity, and diverging colors were used for calibration error. This made the visual language more consistent across the site.

For the supporting charts, we chose to keep the plots simpler and more numerous rather than making each one overloaded. These charts gave users familiar entry points before asking them to interpret the surface. The prototype also introduced interaction through hover details, animated counters, scroll-triggered transitions, and the Markov simulation.

However, the Milestone 2 prototype still had limitations. Some sections were visually polished but not yet fully integrated into a complete narrative. The scrollytelling was mostly limited to fade-in animations rather than meaningful step-by-step changes in the visualization. Some elements, such as the Markov simulation, were interesting but still felt separate from the main calibration story. The final milestone therefore focused on keeping the successful parts of the prototype, simplifying the weaker ones, and making the whole website feel like one coherent experience.

Final Design

The final website is no longer only a set of analysis plots. It is designed as a guided explanation of one central question: *do short-term prediction market prices behave like calibrated forecasts?* Compared with the Milestone 2 prototype, the final version puts more emphasis on narrative structure, visual consistency, and user interaction. We wanted users to understand what the market is, see the evidence for calibration or miscalibration, and then experience the uncertainty of these markets themselves.

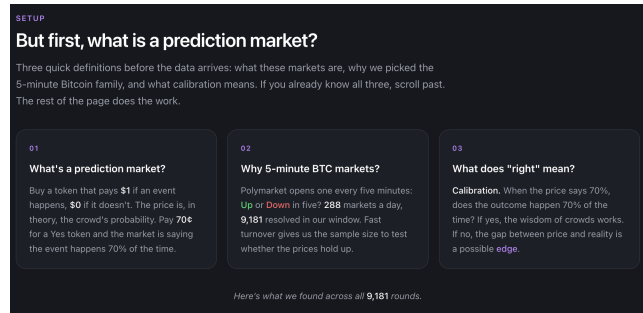


Final landing page introducing the NegativeEV project and the main dataset statistics.

The landing section introduces the project before showing any complex chart. Instead of starting immediately with calibration, it first establishes the visual identity of the website and summarizes the scale of the dataset

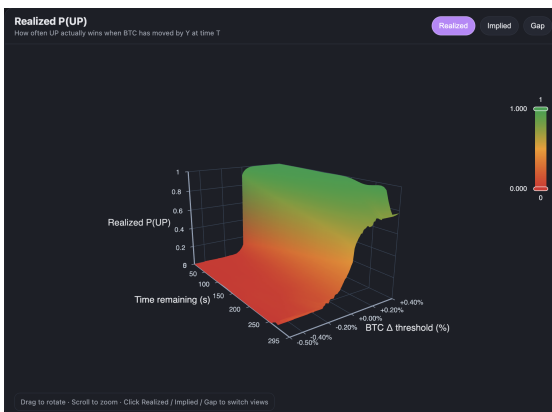
through a small set of headline numbers. This helps users understand that the project is based on thousands of repeated markets, not on a few isolated examples.

A key improvement in the final design is that the website explains the market mechanics before asking users to interpret the analysis. Since prediction markets are not familiar to every viewer, we added explanatory text and visual structure around the idea that an UP or DOWN token price can be read as an implied probability. For example, a token priced at 70 cents corresponds to a market-implied probability of roughly 70%. This explanation makes the later calibration plots easier to understand.

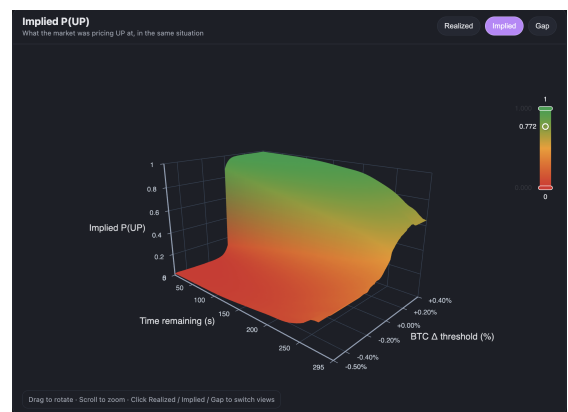


Market explanation section explaining UP/DOWN tokens, market prices, and implied probabilities.

The main analytical component of the final website is the interactive calibration explorer. In the prototype, the 3D surface already showed that calibration depended on time remaining and BTC movement, but it was still difficult to interpret. In the final version, the surface is split into three modes: realized probability, market-implied probability, and calibration gap. This lets users inspect the two ingredients separately before comparing them.

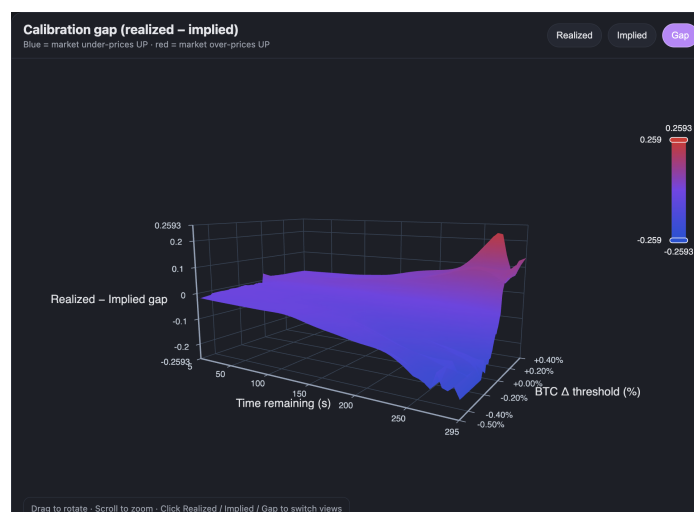


Realized probability mode.



Market-implied probability mode.

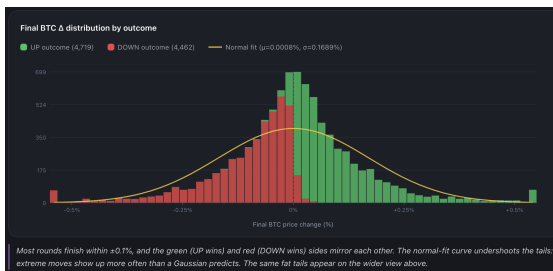
The realized probability view shows what historically happened in similar situations: given the BTC movement and time remaining, how often did the market actually resolve UP? The implied probability view shows what the market was pricing at those same moments. By separating these two surfaces, the website avoids forcing users to interpret the error immediately. They can first build intuition about the historical outcome pattern and the market's belief pattern.



Calibration gap mode, showing where the market appears to over-price or under-price UP outcomes.

The calibration gap view is the most important comparison view. It subtracts the market-implied probability from the realized probability, revealing where the market appears to under-price or over-price UP outcomes. Positive regions suggest that UP happened more often than the market price implied, while negative regions suggest that the market was too optimistic about UP. This view turns the project from a simple probability visualization into an evaluation of market calibration.

We also redesigned the supporting visualizations so that each one answers a specific question. The distribution section explains the underlying BTC movements behind the markets. This is important because most five-minute BTC changes are small, meaning many markets are decided by tiny price movements. The heatmap-style views then show how market behavior changes over time and across market conditions.



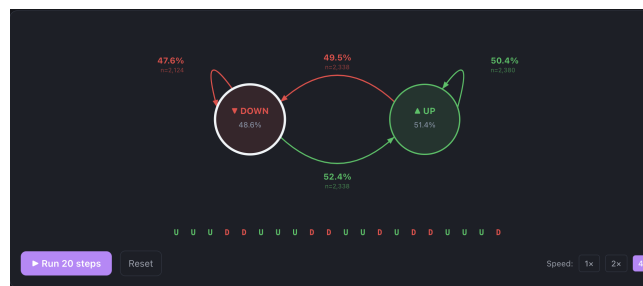
BTC movement distribution.



Temporal or calibration heatmap.

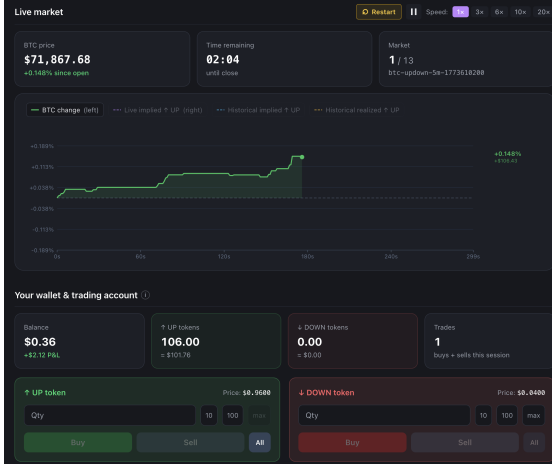
The distribution view gives users a statistical overview of the markets, while the heatmap provides a more structured summary of how outcomes and probabilities vary across time. Together, these charts support the 3D surface without duplicating it. They provide simpler 2D entry points for users who may find the surface harder to read at first.

The final website also keeps the Markov and streak analysis, but it is now positioned as a secondary question rather than a separate story. This section asks whether consecutive market outcomes behave independently, or whether UP and DOWN streaks appear more often than expected. It complements the calibration analysis by showing another way to think about short-term market structure.

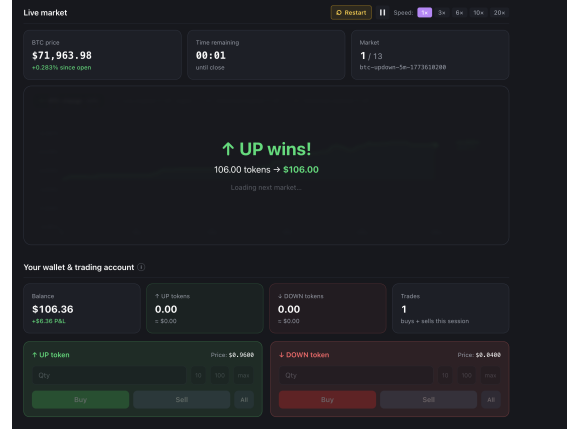


Markov chain and streak analysis showing transitions between consecutive UP and DOWN outcomes.

The largest interactive addition is the trading playground. Calibration charts explain the market from the outside, but the playground lets users experience the decision problem from the inside. It replays historical five-minute market data and allows users to buy or sell simplified UP and DOWN tokens. This makes the website more memorable because users feel how quickly probabilities, prices, and BTC movements can change.



Trading playground before a trade.



Trading playground after a simulated action.

The sandbox is educational rather than financial. It does not provide trading advice; instead, it shows why short-term prediction is difficult. Even when the calibration surface reveals historical patterns, individual markets remain noisy and uncertain. Ending with the sandbox therefore connects the analytical result back to user experience: prediction markets may aggregate information, but acting on that information in real time is still challenging.

Overall, the final design changed the project from a technical dashboard into a guided interactive story. The website first introduces the market and dataset, then explains calibration through the 3D explorer, supports the main findings with focused 2D views, and finally lets users test their intuition in the trading playground. This structure makes the project more accessible for non-experts while still preserving analytical depth for users interested in market efficiency, calibration, and short-term prediction.

Technical Implementation

The final website is built as a reproducible Python data pipeline connected to a React 19 single-page dashboard (Vite, TypeScript, Tailwind CSS 4). The main technical decision was to perform all heavy computation offline: raw Polymarket trades and Binance BTC prices are cleaned, aligned, and aggregated in Python, then exported as compact JSON and standalone HTML artifacts. The processed analytical dataset (`btc_5m_timeseries.parquet`, $\sim 9,181$ markets $\times$ 300 seconds, plus a per-market summary CSV) is committed to the GitHub repository, so reviewers can run the site and regenerate plots without the full ~ 15 GB raw API dumps (`data/raw/` is excluded; reconstruction is documented via `scripts/fetch.btc5m.py` and `scripts/build.dataset.py`).

Raw API JSON $\rightarrow$ Python ETL (`build.dataset.py`) $\rightarrow$ cleaned Parquet/CSV $\rightarrow$ plot scripts (surface + JSON exports) $\rightarrow$ precomputed artifacts $\rightarrow$ React SPA (custom SVG + iframed ECharts-GL)

7.1 Data pipeline

The pipeline is split into small, documented scripts under `scripts/`, with a full reproduction guide in `docs/pipeline.md`:

- **Surface.** `build_cumulative_surface_html.py` computes a directional conditional probability on a grid of time remaining and BTC move thresholds. For $Y \geq 0$: $\mathbb{P}(\text{Up} \mid \Delta_{\text{BTC}} \geq Y, T)$; for $Y < 0$: $\mathbb{P}(\text{Up} \mid \Delta_{\text{BTC}} \leq Y, T)$. The output is `docs/up_cumulative_echarts3d.html`, which contains the three surface modes used in the website: Realized, Implied, and Gap.
- **Website JSON.** `build_analysis_data.py` exports the main frontend assets, including the BTC movement distribution, temporal heatmaps, Markov transitions, daily volume, trades-per-market, volume-vs-change scatter, hero statistics, and 2D calibration curves.
- **Interactivity.** `export_playground_events.py` exports 50 sequential historical markets for the trading replay, while `export_calibration_lookup.py` creates the 2D lookup table used by the live “Historical Prediction Insight” panel in the playground.

7.2 Frontend architecture

The live site (<https://negative-ev.vercel.app>) is a single-scroll narrative with sticky navigation and scroll-spy highlighting. The reading order is: Hero $\rightarrow$ Introduction $\rightarrow$ 3D calibration surface $\rightarrow$ dataset statistics $\rightarrow$ temporal

heatmaps $\rightarrow$ price distributions $\rightarrow$ Markov streaks $\rightarrow$ calibration-over-time curves $\rightarrow$ verdict synthesis $\rightarrow$ trading playground. Each chapter is implemented as a React section under `website/src/components/sections/`, while shared layout elements such as headers, plot frames, and stat cards live in `components/layout/`.

Layer	Role
Python pipeline	Cleans data, computes probabilities, exports Parquet/CSV/JSON/HTML.
ECharts-GL surface	Interactive Realized / Implied / Gap 3D views, embedded from <code>/plots/</code> .
Custom SVG charts	Histograms, heatmaps, scatters, area charts, Markov diagram, and price chart.
React state	Scroll-spy, tooltips, Markov simulation, and playground trading logic.
Calibration lookup	Precomputed 2D table feeding live historical $\mathbb{P}(\text{Up})$ in Full playground mode.

Most two-dimensional views are hand-coded React SVG components with explicit linear scales, axes, color encodings, and hover tooltips. This gave us pixel-level control over the dark trading-platform visual style without relying on a heavy charting framework. Green and red are used consistently for UP and DOWN outcomes, while diverging palettes are used when showing calibration error.

The 3D calibration surface is pre-rendered with Apache ECharts-GL as a self-contained HTML file and embedded into the React site through an iframe. This replaced an earlier Plotly prototype, which caused mesh artifacts and unstable hover behavior on our directional calibration surface.

The trading playground replays real 5-minute markets tick by tick. A reducer manages the user’s balance, open positions, and market progression. The price chart uses a symmetric dynamic y-axis so that future extrema are not leaked to the user. In Full mode, the calibration panel queries the exported lookup table each second to compare live token prices with historical $\mathbb{P}(\text{Up})$ for the current (T, Δ_{BTC}) state.

7.3 Technical challenges

Three issues dominated the implementation. **First, defining the correct calibration metric.** Early versions plotted cumulative or incorrectly conditioned quantities, which mixed incompatible market states and kept probabilities close to the base UP rate of about 51%. The final directional tail definition better matches the trading question: given the current BTC move and time remaining, how often did the market historically finish UP?

Second, keeping the surface and tooltips consistent. The probabilities used for the surface height and the values shown in the tooltip had to come from the same smoothed counts. Otherwise, users could hover over a point and see a value that did not match the visible surface. We therefore computed and exposed the same numerator, denominator, probability, and effective sample size for both the mesh and the tooltip.

Third, preserving performance and visual continuity. The raw dataset contains millions of trades, so the website cannot compute everything in the browser. Pre-aggregating the data into compact Parquet, JSON, and HTML artifacts keeps the single-page app responsive. For the surface, smoothing is applied only along the time axis, while sparse cells are filled along the BTC-threshold direction. This removes holes in the mesh without blurring across the important $Y = 0$ directional boundary.

The repository documents setup, schema, and design rationale in `README.md`, `docs/dataset.md`, `docs/visualizations.md`, and `docs/pipeline.md`. We also included spot-check values, such as $\mathbb{P}(\text{Up} \mid \Delta \geq +0.05\%, T=5\text{ s}) \approx 98.9\%$, to help validate that the regenerated pipeline matches the final website.

AI tools were also used during the project, especially for rapid prototyping, debugging, and iterating on visualization ideas. This supported our workflow by helping us test alternatives quickly while we stayed responsible for the methodology, implementation, and final decisions.

Challenges and Design Decisions

The first major challenge was obtaining a usable dataset. Since Polymarket’s Bitcoin 5-minute markets did not exist as a cleaned research dataset, we had to collect market metadata, trades, and outcomes from the Polymarket API, then align them with Binance BTC spot prices. This required dealing with API limitations, incomplete or inactive markets, timestamp alignment, and a raw dataset that was too large to use directly in the browser. We therefore moved the expensive processing steps offline and exported compact files for the final website.

The second challenge was defining calibration in a way that matched the actual market question. Early plots compared prices and outcomes in simpler ways, but they did not always capture the conditional nature of a live five-minute market. In the final version, the calibration surface focuses on how realized outcomes change with time remaining and BTC price movement, then compares this with the market-implied probability. This made the analysis more meaningful and easier to connect to the user’s experience on the website.

The 3D calibration surface was the hardest visualization to implement. Plotly worked well for prototyping, but it caused issues with hover text, mesh artifacts, and rendering stability. We therefore switched to ECharts-GL and

generated the surface offline in Python before embedding it into the React website. This made the interaction smoother and allowed users to switch between the realized, implied, and gap views without expensive browser-side computation.

A major design decision was to avoid turning the final website into a dense technical dashboard. During exploration, we produced many charts, including distributions, heatmaps, scatter plots, volume charts, Markov transitions, and calibration curves. For the final website, we kept only the views that supported the main story: the landing page introduces the dataset, the market mechanics section explains UP and DOWN tokens, the 3D surface presents the main calibration question, and the supporting charts provide simpler context around BTC movements and market patterns.

The trading playground introduced another important design challenge. We wanted users to experience how difficult short-term prediction feels in practice, but without making the project look like trading advice. The final playground therefore replays historical five-minute markets in a simplified educational setting. Users can buy and sell UP or DOWN tokens, while the full mode connects the simulation back to the calibration analysis through historical prediction insight.

Finally, the visual design had to balance atmosphere and clarity. We chose a dark trading-inspired interface, green and red outcome colors, and bright highlights to match the prediction-market setting. At the same time, we simplified the number of charts, reduced unnecessary jargon, and organized the website as a guided scroll story. Overall, the main design decision was to prioritize clarity over completeness: instead of showing every analysis we produced, the final website presents the parts that best explain whether prediction market prices match reality.

Team Contributions

The project was developed collaboratively, but each team member focused on different parts of the pipeline, visualization, and presentation. Many components were co-owned, especially the final website integration, narrative refinement, and debugging before deployment.

Anton Svet:

Anton was responsible for the raw data acquisition pipeline, gathering and aligning data from both Polymarket and Binance. This included dealing with API limitations, collecting market-level and trade-level information, filtering trades, and ensuring that the different data sources could be connected through consistent timestamps. He also developed the updated directional calibration surface, migrated the prototype from Plotly to ECharts-GL, contributed to the trading sandbox logic, maintained pipeline documentation, and helped integrate the final components into the completed website. Across these tasks, he also ensured that the final data artifacts, website figures, and process book followed the same methodology.

Santiago Rivadeneira:

Santiago focused on extracting insights from the dataset and translating them into visual explanations. He carried out advanced data analysis and implemented several key visual components, including the BTC distribution plots, temporal heatmaps, activity charts, calibration curves, and Markov chain analysis. Beyond the charts themselves, he played a central role in shaping the visual identity and narrative architecture of the website, including the dark theme, color palette, section ordering, animated statistics, and guided scroll structure. His work helped connect the individual visual sections into a coherent story, with simpler contextual charts preparing users for the more complex calibration views.

Arthur Margeat:

Arthur led the dataset cleaning and preprocessing work, filtering empty or anomalous markets and transforming the raw data into a clean structure that could support the rest of the project. He also worked on the first exploratory analyses, which helped the team decide which questions were visually and statistically meaningful. Later, he handled deployment and improved the trading playground by incorporating historical calibration information, lookup-based prediction hints, verdict logic, and end-of-session feedback for users. This also involved validating the final numerical claims so that the deployed version remained consistent with the cleaned dataset and exported analysis files.

Overall, the project evolved from a technical exploration of a new dataset into a guided interactive visualization. The most important team decision was to prioritize clarity over completeness: instead of showing every analysis we produced, we selected the views that best supported the calibration story. If we had more time, we would further improve mobile responsiveness, add more guided annotations to the 3D surface, and expand the sandbox with more user feedback after each simulated market.